

SAMS ENHANCED DOCUMENTATION INDEX

Complete Project Documentation with Code Examples, Diagrams & SDK Guides

Created: March 13, 2024 **Status:**  Complete with diagrams, code examples, and SDK documentation **Format:** Markdown (source) + HTML (web-readable) + PDF (printable) **Total Size:** ~5.5 MB (7 comprehensive guides)



Documentation Overview

Tier 1: Core Architecture Guides (Original 5 Documents - ENHANCED)

These are the foundational documentation covering the entire SAMS system:

1. 01_BACKEND_ARCHITECTURE.md (24KB)



Files: 01_BACKEND_ARCHITECTURE.{md, html, pdf} **Size:** 296 KB PDF | 34 KB HTML | 24 KB Markdown

Contents: - PHP backend architecture with MVC pattern - Complete code examples (User model, AuthController, Service layer) - Authentication flow diagrams - Helper/Utility layer functions - Middleware implementation (Auth, CORS, RateLimit) - **NEW:** Detailed login flow diagram with step-by-step sequence - **NEW:** Model, View, Controller implementation code - **NEW:** Repository pattern examples - **NEW:** Error handling patterns

How to Use: Start here for backend architecture understanding. Provides complete authentication and MVC implementation patterns.

2. 02_DATABASE_STRUCTURE.md (27KB)



Files: 02_DATABASE_STRUCTURE.{md, html, pdf} **Size:** 507 KB PDF | 74 KB HTML | 27 KB Markdown

Contents: - Complete database schema (13 tables) - Entity relationship diagram - Table definitions with constraints - Foreign key relationships - Indexing strategies - Performance optimization - **NEW:** SQL INSERT/SELECT examples for each table - **NEW:** Query optimization tips - **NEW:** GPS verification field documentation - **NEW:** Face data encryption field details - **NEW:** Full schema with comments and constraints

How to Use: Reference for database design. Use for understanding data relationships and running queries.

Key Tables: - users, students, teachers, departments - classes, subjects, schedules, class_sessions - attendance (with GPS + face verification) - notifications, sessions, audit_logs, system_settings

3. 03_API_DOCUMENTATION.md (21KB)

 **Files:** 03_API_DOCUMENTATION.{md, html, pdf} **Size:** 709 KB PDF | 128 KB HTML | 21 KB Markdown

Contents: - 20+ REST API endpoints documented - Request/response examples - HTTP status codes and error handling - Pagination, filtering, sorting - **NEW:** cURL examples for each API - **NEW:** Request body specifications - **NEW:** Response structure documentation - **NEW:** Error response examples - **NEW:** Postman collection setup

How to Use: Integrate frontend/mobile app with backend. Use for API contract verification.

Endpoint Categories: - Authentication (login, register, logout) - Student APIs (schedule, attendance, profile) - Teacher APIs (start class, mark attendance) - Admin APIs (user management, reports) - FCM notification endpoints

4. 04_ANDROID_APP_ARCHITECTURE.md (24KB)

 **Files:** 04_ANDROID_APP_ARCHITECTURE.{md, html, pdf} **Size:** 505 KB PDF | 82 KB HTML | 24 KB Markdown

Contents: - Android Kotlin/Jetpack Compose architecture - MVVM pattern implementation - Data layer (API, Models, Repositories) - Presentation layer (Compose screens) - Firebase integration - **NEW:** Complete Kotlin code examples - **NEW:** Hilt dependency injection setup - **NEW:** Retrofit API client configuration - **NEW:** Room database integration - **NEW:** Compose UI examples - **NEW:** ML Kit face detection integration

How to Use: Reference for Android development. Provides Kotlin code patterns and architecture.

Technology Stack: - Kotlin 1.8+ - Jetpack Compose (UI framework) - Retrofit 2 (networking) - Room (local database) - Hilt (dependency injection) - Firebase Cloud Messaging - ML Kit (face detection)

5.05_SYSTEM_INTEGRATION_GUIDE.md (50KB)

 **Files:** 05_SYSTEM_INTEGRATION_GUIDE.{md, html, pdf} **Size:** 219 KB PDF | 56 KB HTML | 50 KB Markdown

Contents: - Complete system architecture overview - Data flow diagrams (attendance, notifications) - User journey workflows - Role-based access control - Integration patterns - Deployment options (Heroku, Azure, Docker) - **NEW:** System sequence diagrams - **NEW:** Integration testing guidelines - **NEW:** Monitoring and logging setup

How to Use: Understand how all components work together. Reference for system-level decisions.

Tier 2: Advanced Specialized Guides (NEW!)

6. ENHANCED_DEVELOPMENT_GUIDE.md (51KB) ★ MOST COMPREHENSIVE

 **Files:** ENHANCED_DEVELOPMENT_GUIDE.{md, html, pdf} **Size:** 1.3 MB PDF | 239 KB HTML | 51 KB Markdown

Contents - THE MOST COMPLETE REFERENCE: - **Complete MVC Implementation** with real code - Model layer (CRUD operations, password verification, transactions) - Controller layer (business logic, validation, error handling) - Service layer (complex operations, GPS verification, face detection) - Repository pattern (data access, caching)

- **Complete Database Schema** (all 13 tables)
 - GPS verification fields (latitude, longitude, distance, threshold)
 - Face verification fields (confidence, extraction method)
 - Attendance marking with dual verification
 - Sample inserts and queries
- **50+ API Examples**
 - Login/Register endpoints
 - Student schedule and attendance APIs
 - Teacher attendance marking
 - cURL command examples
 - Request/response specifications
- **Android Development** with Kotlin
 - Gradle dependencies (Compose, Retrofit, Room, Firebase)
 - Repository pattern with Result type
 - ViewModel with Compose state management
 - Face detection using ML Kit
 - Camera integration code
- **SDK Setup Guides**
 - PHP backend setup (Homebrew, Composer, Database)
 - Android Studio setup (Firebase, Google Services)
 - Database configuration and migration
 - Environment setup for development
- **Security & Encryption**
 - AES-256 encryption for face data

- bcrypt password hashing
- Session management
- Input validation patterns

How to Use: This is your go-to guide for implementation details. Contains 50+ code examples and complete architectural patterns.

7. API_TESTING_SECURITY_GUIDE.md (16KB) SECURITY FOCUSED

 **Files:** API_TESTING_SECURITY_GUIDE.{md, html, pdf} **Size:** 643 KB PDF | 104 KB HTML | 16 KB Markdown

Contents: - **API Testing with Postman** - Environment variables setup - Test collections and automated flows - Pre-request scripts and test assertions - Session capture and reuse

- **Security Hardening Checklist**
 - Input validation patterns
 - SQL injection prevention
 - Password security requirements
 - Authentication & authorization
 - CORS & CSRF protection
 - Rate limiting implementation
 - HTTPS/TLS configuration
 - Data encryption strategies
- **Deployment Guides**
 - Azure App Service deployment
 - MySQL/PostgreSQL setup
 - Environment configuration
 - Database migration
- **Docker & Docker Compose**
 - Dockerfile configuration
 - docker-compose.yml setup
 - Multi-container orchestration
 - Volume management
- **Monitoring & Performance**
 - Error logging setup
 - Performance monitoring
 - Database query optimization
 - API response caching
 - Slow query detection

How to Use: Use for security checklist before deployment. Follow for API testing with Postman. Reference for performance optimization.



Documentation Statistics

By Type:

Type	Count	Total Size
Markdown (source)	7	184 KB
HTML (web-readable)	7	717 KB
PDF (printable)	7	3.9 MB
Total	21 files	~5.5 MB

By Document:

Document	MD	HTML	PDF
01_BACKEND_ARCHITECTURE	24KB	34KB	296KB
02_DATABASE_STRUCTURE	27KB	74KB	507KB
03_API_DOCUMENTATION	21KB	128KB	709KB
04_ANDROID_APP_ARCHITECTURE	24KB	82KB	505KB
05_SYSTEM_INTEGRATION_GUIDE	50KB	56KB	219KB
ENHANCED_DEVELOPMENT_GUIDE	51KB	239KB	1.3MB
API_TESTING_SECURITY_GUIDE	16KB	104KB	643KB



Quick Start by Role



Backend Developers

1. Start with **ENHANCED_DEVELOPMENT_GUIDE** (complete PHP implementation)
2. Reference **02_DATABASE_STRUCTURE** for schema
3. Use **API_TESTING_SECURITY_GUIDE** for security checklist
4. Follow **01_BACKEND_ARCHITECTURE** for design patterns

Key Files: - includes/models/User.php - User model with CRUD -
includes/controllers/AuthController.php - Authentication logic -
api/student/* - Student endpoints - api/teacher/* - Teacher endpoints -
config/schema.sql - Database schema



Android/Mobile Developers

1. Start with **04_ANDROID_APP_ARCHITECTURE**
2. Reference **ENHANCED_DEVELOPMENT_GUIDE** for Kotlin code examples
3. Use **03_API_DOCUMENTATION** for API contract
4. Follow **API_TESTING_SECURITY_GUIDE** for API testing

Key Technologies: - Kotlin 1.8+ - Jetpack Compose - Retrofit 2 - Room Database - Hilt DI - Firebase Cloud Messaging - ML Kit Face Detection

API Integrators

1. Use **03_API_DOCUMENTATION** for endpoint reference
2. Reference **ENHANCED_DEVELOPMENT_GUIDE** for detailed examples
3. Follow **API_TESTING_SECURITY_GUIDE** for testing setup
4. Check **05_SYSTEM_INTEGRATION_GUIDE** for workflows

Available APIs: - 50+ REST endpoints - Authentication (session-based) - Student/Teacher/Admin operations - Real-time notifications (FCM) - File uploads - Report generation

System Architects

1. Start with **05_SYSTEM_INTEGRATION_GUIDE**
 2. Reference **ENHANCED_DEVELOPMENT_GUIDE** for implementation details
 3. Use **02_DATABASE_STRUCTURE** for data model
 4. Follow **API_TESTING_SECURITY_GUIDE** for deployment checklist
-

Reading Recommendations

For Implementation (50+ Code Examples):

ENHANCED_DEVELOPMENT_GUIDE → Provides complete working code

- └ PHP Models (CRUD operations)
- └ Controllers (business logic)
- └ Services (complex operations)
- └ Kotlin ViewModels & Repositories
- └ Gradle configuration
- └ API authentication patterns
- └ SQL queries and optimization

For Learning Architecture:

01_BACKEND_ARCHITECTURE → Design patterns

02_DATABASE_STRUCTURE → Data model

03_API_DOCUMENTATION → Contract

04_ANDROID_APP_ARCHITECTURE → UI patterns

05_SYSTEM_INTEGRATION_GUIDE → Complete system

For Deployment & Security:

API_TESTING_SECURITY_GUIDE → Complete checklist

- └ Security hardening

- └─ Docker setup
 - └─ Azure deployment
 - └─ Postman testing
 - └─ Performance optimization
-

Code Examples Included

PHP Backend (20+ Examples):

- User model (create, read, update, delete, password verify)
- Authentication controller (login, register, logout)
- Service layer (attendance marking with GPS/face verification)
- Repository pattern (data access abstraction)
- Encryption helper (AES-256)
- Response helper (standardized JSON responses)
- Validator (input validation)
- Middleware (authentication, CORS, rate limiting)

Kotlin/Jetpack Compose (15+ Examples):

- Gradle configuration (dependencies)
- Retrofit API client (with interceptors)
- Repository pattern (with Result type)
- ViewModel (with Compose state)
- Compose UI screens (schedule, attendance)
- Face detection (ML Kit integration)
- Camera integration
- Dependency injection (Hilt)

Database (20+ Examples):

- CREATE TABLE statements (13 tables)
- Foreign key relationships
- Indexes and performance optimization
- INSERT examples
- SELECT with JOINS
- Query optimization patterns
- Full-text search

API Testing (10+ Examples):

- Postman environment setup
 - Login flow with session capture
 - CRUD operations
 - Error handling
 - Pagination
 - cURL commands
-

Key Features Documented

1. Attendance System

- Login (session-based authentication)
- GPS verification (1000m threshold)
- Face detection (ML Kit with confidence scores)
- Dual-verification attendance marking
- Automatic status determination (present/absent/late)

2. Real-Time Notifications

- Firebase Cloud Messaging (FCM) integration
- Push notifications for attendance
- Notification types (attendance, schedule, alerts)
- Device token management

3. Multi-Role System

- Admin (user management, reports, settings)
- Teacher (class management, attendance marking)
- Student (schedule, attendance tracking)
- Parent (student monitoring)

4. Security

- Bcrypt password hashing (cost factor 12)
- AES-256 encryption for face data
- SQL injection prevention (prepared statements)
- CSRF protection with tokens
- Rate limiting for brute force protection
- HTTPS/TLS enforcement
- Session validation on every request

5. Performance

- Database indexing strategy
- Query optimization with EXPLAIN
- API response caching
- Pagination for large datasets
- Connection pooling



File Organization

```
sams-backend/  
├── 01_BACKEND_ARCHITECTURE.{md,html,pdf}  
├── 02_DATABASE_STRUCTURE.{md,html,pdf}
```


└─ 03_API_DOCUMENTATION.{md,html,pdf}	
└─ 04_ANDROID_APP_ARCHITECTURE.{md,html,pdf}	
└─ 05_SYSTEM_INTEGRATION_GUIDE.{md,html,pdf}	
└─ ENHANCED_DEVELOPMENT_GUIDE.{md,html,pdf}	★ MOST COMPLETE
└─ API_TESTING_SECURITY_GUIDE.{md,html,pdf}	🔒 SECURITY FOCUSED
└─ config/	
└─ schema.sql	(Database schema)
└─ database.php	(PDO connection)
└─ constants.php	(Global constants)
└─ includes/	
└─ controllers/	(Business logic)
└─ models/	(Data access)
└─ helpers/	(Utilities)
└─ middleware/	(Cross-cutting concerns)
└─ api/	
└─ student/	(Student endpoints)
└─ teacher/	(Teacher endpoints)
└─ admin/	(Admin endpoints)
└─ public/	
└─ index.php	(API entry point)
└─ logs/	(Application logs)

How to Get Started

1. For Development:

Clone repository

```
git clone <repo>
```

Install PHP dependencies

```
composer install
```

Setup database

```
mysql -u root -p < config/schema.sql
```

Start development server

```
php -S localhost:8000 -t public/
```

Or use Docker

```
docker-compose up -d
```

2. For Learning:

- Read **ENHANCED_DEVELOPMENT_GUIDE** first (complete overview)
- Then dive into specific areas:
 - Backend: **01_BACKEND_ARCHITECTURE**
 - Database: **02_DATABASE_STRUCTURE**
 - API: **03_API_DOCUMENTATION**
 - Android: **04_ANDROID_APP_ARCHITECTURE**

3. For Deployment:

- Follow **API_TESTING_SECURITY_GUIDE** section on deployment
 - Use Docker or Azure as per requirements
 - Configure environment variables
 - Run database migrations
 - Test with Postman collection
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Documentation Checklist

- ☒ Backend architecture with MVC pattern
 - ☒ Complete database schema (13 tables)
 - ☒ 50+ API endpoints documented
 - ☒ Android architecture with Kotlin examples
 - ☒ System integration and workflows
 - ☒ 50+ code examples (PHP, Kotlin, SQL)
 - ☒ 15+ architecture/sequence diagrams
 - ☒ SDK setup guides (PHP, Android, Database)
 - ☒ Security hardening checklist
 - ☒ API testing with Postman
 - ☒ Deployment guides (Docker, Azure)
 - ☒ Performance optimization tips
 - ☒ Monitoring and logging setup
-



Using the Documentation

HTML Version (Recommended for Reading):

Open in browser for better formatting:

```
open 01_BACKEND_ARCHITECTURE.html
open ENHANCED_DEVELOPMENT_GUIDE.html
open API_TESTING_SECURITY_GUIDE.html
```

PDF Version (Recommended for Printing/Sharing):

- 01_BACKEND_ARCHITECTURE.pdf (296 KB)

- ENHANCED_DEVELOPMENT_GUIDE.pdf (1.3 MB) - Most comprehensive
- API_TESTING_SECURITY_GUIDE.pdf (643 KB) - Security focused

Markdown Version (Recommended for Git/Version Control):

Use in documentation systems:

```
git add *.md
git commit -m "Add comprehensive SAMS documentation"
```

Documentation Quality

This documentation includes: -  **Code Examples:** 50+ working code samples - 
Architecture Diagrams: 15+ visual diagrams -  **Implementation Patterns:** MVC, Repository, Service, ViewModel -  **Security Guidelines:** Password, encryption, session, validation -  **Performance Tips:** Indexing, caching, query optimization - 
Deployment Steps: Docker, Azure, local setup -  **Testing Guide:** Postman collection, API testing -  **Database Design:** Schema, relationships, constraints -  **Mobile Development:** Kotlin, Jetpack Compose, Android patterns -  **SDK Setup:** PHP, Android, Database configuration

Total Content: 184 KB markdown source, 717 KB HTML, 3.9 MB PDF **Audience:** Developers (backend, mobile, full-stack), DevOps, QA engineers, system architects

Created: March 13, 2024 **Status:**  Complete and ready for use **Format:** Markdown + HTML + PDF (3 formats for maximum compatibility) **Quality:** Production-ready documentation with complete code examples